Symptom Clusters in Cancer (PDQ®)–Health Professional Version

Overview

During and after treatment for cancer, patients experience an average of 10 to14 co-occurring symptoms.[1,2] Compared with patients with few or no symptoms, patients with multiple concurrent symptoms may experience worse decrements in physical and cognitive function,[3-6] quality of life,[3-6] and overall survival.[7] Symptoms that occur together over time may form a cluster.

A **symptom cluster** is two or more concurrent symptoms with the following properties:[1,8]

- The concurrent symptoms within the cluster are stable (i.e., the same symptoms that comprise clusters are identified at different time points throughout the cancer care trajectory).
- The symptoms within a cluster may share underlying mechanisms.
- The symptoms within a cluster may share outcomes.
- The symptom cluster may have a temporal dimension (i.e., the severity of the symptoms within the cluster may shift together over time).

Questions?

Symptoms in a cluster may be related to each other in multiple ways. They may share a common set of biological, psychological, or social mechanisms, or they may cause or potentiate one another's effects.[8] They may also be due to side effects of another treatment. The management of symptom clusters primarily involves identifying relationships among the symptoms. This approach supports strategies that may address multiple symptoms simultaneously. It also avoids unnecessary interventions that can worsen symptom burden and the patient's quality of life. As advancements in symptom science research are achieved, better therapeutic targets will be identified for improved management of symptom clusters and outcomes.

References

1. Miaskowski C, Barsevick A, Berger A, et al.: Advancing Symptom Science Through Symptom Cluster Research: Expert Panel Proceedings and Recommendations. J Natl Cancer Inst 109 (4): , 2017. [PUBMED Abstract]
2. Harris CS, Dodd M, Kober KM, et al.: Advances in Conceptual and Methodological Issues in Symptom Cluster Research: A 20-Year Perspective. ANS Adv Nurs Sci 45 (4): 309-322, 2022 Oct-Dec 01. [PUBMED Abstract]

3. Dodd MJ, Cho MH, Cooper BA, et al.: Identification of latent classes in patients who are receiving biotherapy based on symptom experience and its effect on functional status and quality of life. Oncol Nurs Forum 38 (1): 33-42, 2011. [PUBMED Abstract]
4. Miaskowski C, Dunn L, Ritchie C, et al.: Latent Class Analysis Reveals Distinct Subgroups of Patients Based on Symptom Occurrence and Demographic and Clinical Characteristics. J Pain Symptom Manage 50 (1): 28-37, 2015. [PUBMED Abstract]
5. Neijenhuijs KI, Peeters CFW, van Weert H, et al.: Symptom clusters among cancer survivors: what can machine learning techniques tell us? BMC Med Res Methodol 21 (1): 166, 2021. [PUBMED Abstract]
6. Trudel-Fitzgerald C, Savard J, Ivers H: Longitudinal changes in clusters of cancer patients over an 18-month period. Health Psychol 33 (9): 1012-22, 2014. [PUBMED Abstract]
7. Kerrigan K, Patel SB, Haaland B, et al.: Prognostic Significance of Patient-Reported Outcomes in Cancer. JCO Oncol Pract 16 (4): e313-e323, 2020. [PUBMED Abstract]
8. Aktas A: Cancer symptom clusters: current concepts and controversies. Curr Opin Support Palliat Care 7 (1): 38-44, 2013. [PUBMED Abstract]

Identification of Symptom Clusters: Methods and Limitations

To identify cancer-related symptom clusters, investigators ask patients to complete valid, reliable symptom questionnaires.[1] Symptom data can also be obtained directly from clinicians and/or medical records. Symptom data are analyzed to identify symptoms that occur in clusters, using methods such as exploratory factor analysis (EFA).[2,3] The objective of an EFA is to identify groups of symptoms that are correlated with each other. These groups are known as **factors** or, in the field of symptom cluster research, **symptom clusters**. Factor analysis is consistent with a biopsychosocial theory [4] that the underlying mechanisms of symptoms are likely shared within symptom clusters.

Another method for identifying symptom clusters is through structural equation modeling (SEM).[3] Compared with factor analysis that retrospectively identifies clusters, SEM requires an *a priori* hypothesis about a common cause of the symptom cluster.[3] One benefit of SEM is that direct and indirect, as well as latent, associations between factors can be identified.[3] This method can help identify risk factors in future patients.

Once a symptom cluster has been identified, investigators perform additional analyses to describe patients' experiences with a symptom cluster. A person-centered analytic approach, such as latent class profile analysis, is used to identify and describe subgroups of patients who are similarly burdened by a symptom cluster.[5,6] The symptom cluster burden has multiple dimensions, including the occurrence, frequency, severity, and distress associated with the symptom cluster.[6,7] Determining which patients are at risk for the highest level of symptom cluster burden and associated underlying mechanisms are ongoing areas of research.

Common characteristics associated with a higher symptom cluster burden include the following:[1,8-10]

- Younger age.
- Non-White race and ethnicity.
- Unmarried/unpartnered.
- Unemployed.
- Lower annual income.
- Child care responsibilities.
- Higher comorbidity burden.
- Lower functional status.

One limitation in identifying and managing symptom clusters is that symptom assessment tools and analytic approaches may vary across clinical practices. However, the consistency of symptom clusters identified across studies using different instruments supports the fact that core symptoms are being captured and that underlying mechanisms may be associated with symptoms within a cluster. For example, two symptom clusters common to multiple cancer diagnoses and treatment regimens are a sickness behavior syndrome (fatigue, sleep disturbance, lethargy, depression, loss of appetite) [11-13] and a psychoneurological symptom cluster (depression, cognitive disturbance, sleep disturbance, fatigue, pain).[14,15] Both of these symptom clusters are associated with underlying inflammatory mechanisms.[11,12,15]

Another limitation is that while some symptom clusters appear to be stable over time, [16,17] a patient's symptom experience may vary widely.[7] Specifically, the occurrence of symptoms within a cluster, the level of symptom cluster severity, and the level of distress associated with one or more symptom clusters may shift over time in each individual. Symptom clusters can also persist long after treatment.[7,18]

Due to the complexity of symptom clusters, the symptoms within each cluster, and the temporal nature of symptoms/clusters that may shift over time, it is important to continually assess symptoms from diagnosis through the end of life. Retesting symptom clusters through repeated data-driven analyses (e.g., factor analysis,[2,3] SEM,[3]) can help health care professionals to better understand the stability and/or changes in symptom cluster experiences over time. In addition, symptom clusters may differ between individuals who are undergoing active treatment and those who have completed primary therapy.

For more information about individual side effects, see the summaries on Supportive and Palliative Care.

References

1. Miaskowski C, Cooper BA, Paul SM, et al.: Subgroups of patients with cancer with different symptom experiences and quality-of-life outcomes: a cluster analysis. Oncol Nurs Forum 33 (5): E79-89, 2006. [PUBMED Abstract]

2. Skerman HM, Yates PM, Battistutta D: Identification of cancer-related symptom clusters: an empirical comparison of exploratory factor analysis methods. J Pain Symptom Manage 44 (1): 10-22, 2012. [PUBMED Abstract]
3. Olson K, Hayduk L, Thomas J: Comparing two approaches for studying symptom clusters: factor analysis and structural equation modeling. Support Care Cancer 22 (1): 153-61, 2014. [PUBMED Abstract]
4. Borrell-Carrió F, Suchman AL, Epstein RM: The biopsychosocial model 25 years later: principles, practice, and scientific inquiry. Ann Fam Med 2 (6): 576-82, 2004. [PUBMED Abstract]
5. Miaskowski C, Dunn L, Ritchie C, et al.: Latent Class Analysis Reveals Distinct Subgroups of Patients Based on Symptom Occurrence and Demographic and Clinical Characteristics. J Pain Symptom Manage 50 (1): 28-37, 2015. [PUBMED Abstract]
6. Miaskowski C: Future Directions in Symptom Cluster Research. Semin Oncol Nurs 32 (4): 405-415, 2016. [PUBMED Abstract]
7. Miaskowski C, Barsevick A, Berger A, et al.: Advancing Symptom Science Through Symptom Cluster Research: Expert Panel Proceedings and Recommendations. J Natl Cancer Inst 109 (4): , 2017. [PUBMED Abstract]
8. Doong SH, Dhruva A, Dunn LB, et al.: Associations between cytokine genes and a symptom cluster of pain, fatigue, sleep disturbance, and depression in patients prior to breast cancer surgery. Biol Res Nurs 17 (3): 237-47, 2015. [PUBMED Abstract]
9. Hammer MJ, Cooper B, Paul SM, et al.: Identification of Distinct Symptom Profiles in Cancer Patients Using a Pre-Specified Symptom Cluster. J Pain Symptom Manage 64 (1): 17-27, 2022. [PUBMED Abstract]
10. Pud D, Ben Ami S, Cooper BA, et al.: The symptom experience of oncology outpatients has a different impact on quality-of-life outcomes. J Pain Symptom Manage 35 (2): 162-70, 2008. [PUBMED Abstract]
11. Aktas A: Cancer symptom clusters: current concepts and controversies. Curr Opin Support Palliat Care 7 (1): 38-44, 2013. [PUBMED Abstract]
12. Chan CW, Lee PH, Molassiotis A, et al.: Symptom Clusters in Postchemotherapy Neutropenic Fever in Hematological Malignancy: Associations Among Sickness Behavior Symptom Cluster, Inflammatory Biomarkers of Procalcitonin and C-Reactive Protein, and Febrile Measures. J Pain Symptom Manage 59 (6): 1204-1211, 2020. [PUBMED Abstract]
13. Wood LJ, Weymann K: Inflammation and neural signaling: etiologic mechanisms of the cancer treatment-related symptom cluster. Curr Opin Support Palliat Care 7 (1): 54-9, 2013. [PUBMED Abstract]
14. Trudel-Fitzgerald C, Savard J, Ivers H: Longitudinal changes in clusters of cancer patients over an 18-month period. Health Psychol 33 (9): 1012-22, 2014. [PUBMED Abstract]
15. Kim HJ, Barsevick AM, Fang CY, et al.: Common biological pathways underlying the psychoneurological symptom cluster in cancer patients. Cancer Nurs 35 (6): E1-E20, 2012. [PUBMED

Abstract]

16. Kim E, Jahan T, Aouizerat BE, et al.: Changes in symptom clusters in patients undergoing radiation therapy. Support Care Cancer 17 (11): 1383-91, 2009. [PUBMED Abstract]
17. Pozzar RA, Hammer MJ, Cooper BA, et al.: Stability of Symptom Clusters in Patients With Gynecologic Cancer Receiving Chemotherapy. Cancer Nurs 45 (4): E706-E718, 2022 Jul-Aug 01. [PUBMED Abstract]
18. Brant JM, Beck S, Dudley WN, et al.: Symptom trajectories in posttreatment cancer survivors. Cancer Nurs 34 (1): 67-77, 2011. [PUBMED Abstract]

Predictors and Outcomes of Select Symptom Clusters

Symptom cluster names are often based on symptoms of the cluster (e.g., psychoneurological[1] and gastrointestinal[2]), although names are not standardized across studies. One frequent unnamed cluster identified across cancer diagnoses includes pain, fatigue, sleep disturbance, and depression.[3] This general symptom cluster has been identified in patients with breast,[4,5] lung,[5,6] gastrointestinal,[5,7] and gynecologic [5,8] cancers. Symptom clusters can also reflect underlying disease (e.g., dyspnea, cough, and fatigue in patients with lung cancer) [9] and/or treatments (e.g., sensory neuropathy symptom cluster, mixed motor-sensory symptom cluster related to chemotherapy-induced peripheral neuropathy).[10]

References

1. Kim HJ, Barsevick AM, Fang CY, et al.: Common biological pathways underlying the psychoneurological symptom cluster in cancer patients. Cancer Nurs 35 (6): E1-E20, 2012. [PUBMED Abstract]
2. Harris CS, Miaskowski CA, Conley YP, et al.: Gastrointestinal Symptom Cluster is Associated With Epigenetic Regulation of Lymphotoxin Beta in Oncology Patients Receiving Chemotherapy. Biol Res Nurs 25 (1): 51-64, 2023. [PUBMED Abstract]
3. Trudel-Fitzgerald C, Savard J, Ivers H: Longitudinal changes in clusters of cancer patients over an 18-month period. Health Psychol 33 (9): 1012-22, 2014. [PUBMED Abstract]
4. Doong SH, Dhruva A, Dunn LB, et al.: Associations between cytokine genes and a symptom cluster of pain, fatigue, sleep disturbance, and depression in patients prior to breast cancer surgery. Biol Res Nurs 17 (3): 237-47, 2015. [PUBMED Abstract]
5. Miaskowski C, Cooper BA, Paul SM, et al.: Subgroups of patients with cancer with different symptom experiences and quality-of-life outcomes: a cluster analysis. Oncol Nurs Forum 33 (5): E79-89, 2006. [PUBMED Abstract]
6. Brant JM, Beck S, Dudley WN, et al.: Symptom trajectories in posttreatment cancer survivors. Cancer Nurs 34 (1): 67-77, 2011. [PUBMED Abstract]

7. Lin Y, Bailey DE, Docherty SL, et al.: Distinct morning and evening fatigue profiles in gastrointestinal cancer during chemotherapy. BMJ Support Palliat Care 13 (e2): e373-e381, 2023. [PUBMED Abstract]
8. Pozzar RA, Hammer MJ, Cooper BA, et al.: Symptom Clusters in Patients With Gynecologic Cancer Receiving Chemotherapy. Oncol Nurs Forum 48 (4): 441-452, 2021. [PUBMED Abstract]
9. Cheville AL, Novotny PJ, Sloan JA, et al.: Fatigue, dyspnea, and cough comprise a persistent symptom cluster up to five years after diagnosis with lung cancer. J Pain Symptom Manage 42 (2): 202-12, 2011. [PUBMED Abstract]
10. Wang M, Cheng HL, Lopez V, et al.: Redefining chemotherapy-induced peripheral neuropathy through symptom cluster analysis and patient-reported outcome data over time. BMC Cancer 19 (1): 1151, 2019. [PUBMED Abstract]

Symptom Clusters Common Across Heterogeneous Treatments

Several symptom clusters have been identified across cancer diagnoses and treatments in patients undergoing treatment and throughout survivorship. To date, most studies of symptom clusters in patients undergoing active treatment have analyzed data from patients receiving the same treatment. A limited number of studies suggest the same symptom clusters may occur in groups of patients receiving heterogeneous therapies. A pooled analysis included data from 1,561 patients with advanced cancer in 15 countries. Four symptom clusters were common in patients with myeloma and breast, colorectal, lung, and prostate cancer who were receiving chemotherapy, radiation therapy, analgesics, surgery, and other or no treatments.[1] These clusters included an emotional cluster comprised of feeling tense, worried, irritable, and depressed; a fatigue-pain cluster; a nausea-vomiting cluster; and a cognitive cluster comprised of difficulties with concentration and memory.

In another study of 93 Chinese patients with cervical cancer, mood-cognitive, gastrointestinal, sickness behavior, and pain-related symptom clusters were identified in patients receiving chemoradiation therapy, radiation therapy alone, and surgery followed by chemoradiation. Mean symptom severity scores were not significantly different across these three treatment groups.[2]

Finally, a longitudinal study of 92 patients with metastatic lung cancer assessed interrelationships between depressive symptoms and a sickness behavior symptom cluster. In this sample, 40.5% of participants were receiving chemotherapy, 35.7% were receiving immunotherapy, and 23.8% were receiving oral targeted therapies. All 12 sickness behavior symptoms were significantly associated with patients' scores on the Patient Health Questionnaire-9 and Hospital Anxiety and Depression Scale depression subscale. Sickness behavior symptom severity was significantly higher in patients receiving chemotherapy than in those receiving oral targeted therapies. However, symptom severity among patients receiving immunotherapy did not differ significantly from symptom severity of the other two groups.[3]

In individuals who have completed primary therapy, psychological/cognitive/fatigue clusters are common (e.g., emotional distress, fatigue, cognitive dysfunction;[4] fatigue, difficulty concentrating, drowsiness[5]). In addition, there is evidence for long-term effects in a sickness behavior cluster (e.g., lethargy, sleepiness, depressed mood, hyperalgesia).[6]

In a study of 300 patients receiving palliative chemotherapy, one of the symptom clusters identified was fatigue-cognitive, which was associated with decreased functional status and poor quality of life.[5] The authors highlighted that fatigue is often included in other clusters (e.g., fatigue, dyspnea, drowsiness, pain). Fatigue has also been identified in sickness behavior clusters.[6] Because fatigue is prevalent across a number of symptom clusters, it may be helpful to target fatigue to mitigate the effects of symptom clusters across patient populations.

Additional research using a person-centered approach is needed to understand the individual experiences of symptom clusters over time. Interventions to mitigate a high symptom burden should be tailored to individual symptom cluster profiles.

References

1. Dong ST, Costa DS, Butow PN, et al.: Symptom Clusters in Advanced Cancer Patients: An Empirical Comparison of Statistical Methods and the Impact on Quality of Life. J Pain Symptom Manage 51 (1): 88-98, 2016. [PUBMED Abstract]
2. Wang CL, Wu WY, Lou HM, et al.: Analysis of symptom clusters in Chinese cervical cancer patients undergoing radiotherapy, chemoradiotherapy, or postoperative chemoradiotherapy. Eur J Gynaecol Oncol 38 (3): 398-403, 2017. [PUBMED Abstract]
3. McFarland DC, Walsh LE, Saracino R, et al.: The Sickness Behavior Inventory-Revised: Sickness behavior and its associations with depression and inflammation in patients with metastatic lung cancer. Palliat Support Care 19 (3): 312-321, 2021. [PUBMED Abstract]
4. Levkovich I, Cohen M, Alon S, et al.: Symptom cluster of emotional distress, fatigue and cognitive difficulties among young and older breast cancer survivors: The mediating role of subjective stress. J Geriatr Oncol 9 (5): 469-475, 2018. [PUBMED Abstract]
5. Rha SY, Lee J: Symptom clusters during palliative chemotherapy and their influence on functioning and quality of life. Support Care Cancer 25 (5): 1519-1527, 2017. [PUBMED Abstract]
6. Zick SM, Zwickey H, Wood L, et al.: Preliminary differences in peripheral immune markers and brain metabolites between fatigued and non-fatigued breast cancer survivors: a pilot study. Brain Imaging Behav 8 (4): 506-16, 2014. [PUBMED Abstract]

Pathogenesis of Symptom Clusters

The pathogenesis of symptom clusters has not been clearly established. However, some mechanisms (e.g., chronic inflammation) and interaction effects with patient-specific characteristics (e.g., clinical,

environmental, behavioral) are beginning to unfold. Examples of evidence for common pathogenic mechanisms for selected symptom clusters are summarized below.

One or more symptoms associated with the sickness behavior syndrome (e.g., fatigue, sleep disturbance, lethargy, depression, loss of appetite) are common in a number of other symptom clusters.[1-3] It is a plausible model for understanding the pathogenesis of symptom clusters. The physiological and psychological stress of cancer may be an underlying trigger for the co-occurring symptoms identified in symptom clusters.[4] Central nervous system (CNS) activation of acute psychological stress engages the hypothalamic-pituitary-adrenal axis and sympathetic-adrenal-medullary system to produce increased levels of cortisol. Prolonged stress dampens the homeostatic circadian rhythm of cortisol, causing glucocorticoid resistance, in which immune cells become less sensitive to glucocorticoids.[5] Decreased sensitivity triggers increased levels of glucocorticoids [4,5] as well as epinephrine and norepinephrine.[4]

In addition, cellular responses to regain homeostasis, termed allostasis, include secreting catecholamines and cytokines.[5] With chronic stress, allostatic overload occurs with an increase in pro-inflammatory cytokines. In particular, interleukin (IL)-1-beta, IL-6, tumor necrosis factor-alpha (TNF-alpha), and others are secreted.[3-5] Prolonged secretion of IL-1-beta is associated with sickness behavior symptoms through inducing prostaglandins and activating T and B cells and antibodies.[3,5] In addition, IL-6,[6-8] TNF-alpha,[9-11], and nuclear-factor kappa B [12]—a transcription factor that induces multiple inflammatory cytokines to enter into circulation—have been associated with high symptom severity.

Given the psychological influence of CNS-triggered downstream responses that lead to chronic inflammatory-associated symptoms, individual coping mechanisms and levels of resilience to stress may explain some of the interindividual variability in symptom cluster experiences. Research is just beginning to capture the multiple social determinants of health (e.g., social, structural, environmental, behavioral) that may influence an individual's perception of and response to stress. To best understand the pathogenesis associated with the symptom cluster phenotype, a person-centered approach to research and clinical care is warranted.

References

1. Aktas A: Cancer symptom clusters: current concepts and controversies. Curr Opin Support Palliat Care 7 (1): 38-44, 2013. [PUBMED Abstract]
2. Chan CW, Lee PH, Molassiotis A, et al.: Symptom Clusters in Postchemotherapy Neutropenic Fever in Hematological Malignancy: Associations Among Sickness Behavior Symptom Cluster, Inflammatory Biomarkers of Procalcitonin and C-Reactive Protein, and Febrile Measures. J Pain Symptom Manage 59 (6): 1204-1211, 2020. [PUBMED Abstract]
3. Wood LJ, Weymann K: Inflammation and neural signaling: etiologic mechanisms of the cancer treatment-related symptom cluster. Curr Opin Support Palliat Care 7 (1): 54-9, 2013. [PUBMED Abstract]
4. Cui B, Peng F, Lu J, et al.: Cancer and stress: NextGen strategies. Brain Behav Immun 93: 368-383, 2021. [PUBMED Abstract]

5. Zefferino R, Di Gioia S, Conese M: Molecular links between endocrine, nervous and immune system during chronic stress. Brain Behav 11 (2): e01960, 2021. [PUBMED Abstract]
6. Miaskowski C, Dodd M, Lee K, et al.: Preliminary evidence of an association between a functional interleukin-6 polymorphism and fatigue and sleep disturbance in oncology patients and their family caregivers. J Pain Symptom Manage 40 (4): 531-44, 2010. [PUBMED Abstract]
7. Hiensch AE, Mijwel S, Bargiela D, et al.: Inflammation Mediates Exercise Effects on Fatigue in Patients with Breast Cancer. Med Sci Sports Exerc 53 (3): 496-504, 2021. [PUBMED Abstract]
8. Lengacher CA, Reich RR, Paterson CL, et al.: A Large Randomized Trial: Effects of Mindfulness-Based Stress Reduction (MBSR) for Breast Cancer (BC) Survivors on Salivary Cortisol and IL-6. Biol Res Nurs 21 (1): 39-49, 2019. [PUBMED Abstract]
9. Calvo-Schimmel A, Kober KM, Paul SM, et al.: Sleep disturbance is associated with perturbations in immune-inflammatory pathways in oncology outpatients undergoing chemotherapy. Sleep Med 101: 305-315, 2023. [PUBMED Abstract]
10. Dhruva A, Aouizerat BE, Cooper B, et al.: Cytokine gene associations with self-report ratings of morning and evening fatigue in oncology patients and their family caregivers. Biol Res Nurs 17 (2): 175-84, 2015. [PUBMED Abstract]
11. Kober KM, Harris C, Conley YP, et al.: Perturbations in common and distinct inflammatory pathways associated with morning and evening fatigue in outpatients receiving chemotherapy. Cancer Med 12 (6): 7369-7380, 2023. [PUBMED Abstract]
12. Kober KM, Smoot B, Paul SM, et al.: Polymorphisms in Cytokine Genes Are Associated With Higher Levels of Fatigue and Lower Levels of Energy in Women After Breast Cancer Surgery. J Pain Symptom Manage 52 (5): 695-708.e4, 2016. [PUBMED Abstract]

Symptom Clusters by Treatment Modality

Cancerous tissue and treatment-related side effects are common causes of symptoms within a cluster. Of note, disease and treatments are not always directly related to symptom cluster experiences.[1] In a study of 582 patients receiving treatment for heterogenous types of cancer, the type of cancer diagnosis and treatment received did not predict symptom severity subgroups.[1] This finding suggests that multiple factors likely influence symptom cluster experiences. However, based on known contributors to some symptoms by the type of treatment (e.g., taxane-induced neuropathies [2]), the discussion below highlights symptom clusters identified in patients receiving specific therapies.

Chemotherapy

A substantial body of research has focused on the symptom experiences of people undergoing chemotherapy. To date, symptom clusters have been identified in patients undergoing chemotherapy for brain,[3] breast,[4-11] gastrointestinal,[12,13] gynecologic,[14,15] hematological,[16-19] and lung cancers.

[20,21] Symptom clusters have also been identified in samples of patients undergoing chemotherapy for heterogenous cancer types.[22,23]

In a 2022 systematic review of studies of symptom clusters in patients undergoing chemotherapy, several common symptom clusters were identified across cancer diagnoses. The most common symptom cluster was comprised of psychological symptoms. “Feeling nervous” (or “feeling anxious”) and “feeling sad” (or “depressed mood”) were the most common symptoms in this cluster. Other common symptoms in this cluster included “feeling irritable,” “worrying,” and “difficulty concentrating.” The second most common symptom cluster involved gastrointestinal symptoms, including nausea and abdominal cramps. Additional symptom clusters identified across diagnoses included the following:[24]

- A nutrition/weight-change symptom cluster, with symptoms such as weight gain, weight loss, decreased appetite, and increased appetite.
- An epithelial/body-image symptom cluster, comprised of itching, hair loss, and skin change.
- A respiratory symptom cluster, comprised of shortness of breath, difficulty breathing, and cough.

Certain symptom clusters are common across a limited number of cancer diagnoses. For example, a systematic review found that a vasomotor-hormonal symptom cluster was identified in studies of patients receiving chemotherapy for breast and gynecologic cancers.[24]

Radiation Therapy

Symptom clusters have been described in patients receiving radiation therapy (RT) for brain,[25] breast,[26,27] gastrointestinal,[28] head and neck,[29,30,30] and prostate cancers.[31,32] Other studies have found symptoms clusters in patients receiving RT for a number of cancers.[33,34]

In a study of 1,224 patients with breast cancer receiving RT, symptoms found to cluster together before, during, and after RT included depression and anxiety, nausea and appetite loss, pain and tiredness, and drowsiness, dyspnea, and tiredness.[26] A study of 100 patients with head and neck cancer in Taiwan who were undergoing postoperative RT identified two symptom clusters that were stable throughout treatment: a head and neck–specific cluster (including pain, dry mouth, appetite, sleep disturbance, fatigue, drowsiness, distress, and sadness) and a gastrointestinal cluster (including nausea, vomiting, numbness, dyspnea, and difficulty with memory).[35] In 843 patients undergoing RT for prostate cancer, seven symptom clusters were identified. Early symptoms, older age, and decreased physical function were associated with the severity of late symptom clusters.[32]

In addition, polymorphisms in cytokine genes were associated with symptom clusters in a sample of 157 patients receiving RT for breast and prostate cancer, suggesting that common symptom clusters may be related to inflammatory processes.[36] Polymorphisms in neurotransmitter genes were associated with the severity of sickness behavior, mood-cognitive, and treatment-related symptom clusters, suggesting that mechanisms other than inflammation may also contribute to the development of symptom clusters.[37]

Immunotherapy

Immunotherapies are a relatively new approach to cancer treatments, and numerous newly developed immunotherapy treatments are in phase I clinical trials. As a result, the scope of symptom clusters in patients receiving these therapies is yet to be understood.

References

1. Miaskowski C, Cooper BA, Melisko M, et al.: Disease and treatment characteristics do not predict symptom occurrence profiles in oncology outpatients receiving chemotherapy. Cancer 120 (15): 2371-8, 2014. [PUBMED Abstract]
2. Wang M, Cheng HL, Lopez V, et al.: Redefining chemotherapy-induced peripheral neuropathy through symptom cluster analysis and patient-reported outcome data over time. BMC Cancer 19 (1): 1151, 2019. [PUBMED Abstract]
3. Kim S: A Longitudinal Study of Lipid Peroxidation and Symptom Clusters in Patients With Brain Cancers. Nurs Res 67 (5): 387-394, 2018. [PUBMED Abstract]
4. Albusoul RM, Berger AM, Gay CL, et al.: Symptom Clusters Change Over Time in Women Receiving Adjuvant Chemotherapy for Breast Cancer. J Pain Symptom Manage 53 (5): 880-886, 2017. [PUBMED Abstract]
5. Berger AM, Kumar G, LeVan TD, et al.: Symptom Clusters and Quality of Life over 1 Year in Breast Cancer Patients Receiving Adjuvant Chemotherapy. Asia Pac J Oncol Nurs 7 (2): 134-140, 2020. [PUBMED Abstract]
6. Browall M, Brandberg Y, Nasic S, et al.: A prospective exploration of symptom burden clusters in women with breast cancer during chemotherapy treatment. Support Care Cancer 25 (5): 1423-1429, 2017. [PUBMED Abstract]
7. Li H, Sereika SM, Marsland AL, et al.: Symptom Clusters in Women With Breast Cancer During the First 18 Months of Adjuvant Therapy. J Pain Symptom Manage 59 (2): 233-241, 2020. [PUBMED Abstract]
8. Sullivan CW, Leutwyler H, Dunn LB, et al.: Stability of Symptom Clusters in Patients With Breast Cancer Receiving Chemotherapy. J Pain Symptom Manage 55 (1): 39-55, 2018. [PUBMED Abstract]
9. Wiggenraad F, Bolam KA, Mijwel S, et al.: Long-Term Favorable Effects of Physical Exercise on Burdensome Symptoms in the OptiTrain Breast Cancer Randomized Controlled Trial. Integr Cancer Ther 19: 1534735420905003, 2020. [PUBMED Abstract]
10. Vuttanon N, Finnegan L, Lojanapiwat B, et al.: Effect of progressive muscle relaxation on symptom clusters in breast cancer patients receiving chemotherapy: A quasi-experimental controlled trial. Complement Ther Clin Pract 37: 27-31, 2019. [PUBMED Abstract]
11. Li H, Sereika SM, Marsland AL, et al.: Impact of chemotherapy on symptoms and symptom clusters in postmenopausal women with breast cancer prior to aromatase inhibitor therapy. J Clin Nurs 28 (23-24): 4560-4571, 2019. [PUBMED Abstract]

12. Han CJ, Reding K, Cooper BA, et al.: Stability of Symptom Clusters in Patients With Gastrointestinal Cancers Receiving Chemotherapy. J Pain Symptom Manage 58 (6): 989-1001.e10, 2019. [PUBMED Abstract]

13. Han CJ, Reding K, Cooper BA, et al.: Symptom Clusters in Patients With Gastrointestinal Cancers Using Different Dimensions of the Symptom Experience. J Pain Symptom Manage 58 (2): 224-234, 2019. [PUBMED Abstract]

14. Pozzar RA, Hammer MJ, Cooper BA, et al.: Symptom Clusters in Patients With Gynecologic Cancer Receiving Chemotherapy. Oncol Nurs Forum 48 (4): 441-452, 2021. [PUBMED Abstract]

15. Pozzar RA, Hammer MJ, Cooper BA, et al.: Stability of Symptom Clusters in Patients With Gynecologic Cancer Receiving Chemotherapy. Cancer Nurs 45 (4): E706-E718, 2022 Jul-Aug 01. [PUBMED Abstract]

16. Lin DM, Yin XX, Wang N, et al.: Consensus in Identification and Stability of Symptom Clusters Using Different Symptom Dimensions in Newly Diagnosed Acute Myeloid Leukemia Patients Undergoing Induction Therapy. J Pain Symptom Manage 57 (4): 783-792, 2019. [PUBMED Abstract]

17. Chen F, Leng Y, Zhang L, et al.: The Correlation of Symptom Clusters and Functional Performance in Adult Acute Leukemia Patients Under Chemotherapy. Cancer Nurs 44 (5): E287-E295, 2021 Sep-Oct 01. [PUBMED Abstract]

18. Cherwin CH, Perkhounkova Y: Distress-Based Gastrointestinal Symptom Clusters and Impact on Symptom Interference and Quality of Life in Patients with a Hematologic Malignancy Receiving Chemotherapy. J Pain Symptom Manage 53 (4): 751-758, 2017. [PUBMED Abstract]

19. Sezgin MG, Bektaş H: Symptom Clustering and Its Effect on Functional Status in Lymphoma Patients. Florence Nightingale J Nurs 28 (2): 143-154, 2020. [PUBMED Abstract]

20. Russell J, Wong ML, Mackin L, et al.: Stability of Symptom Clusters in Patients With Lung Cancer Receiving Chemotherapy. J Pain Symptom Manage 57 (5): 909-922, 2019. [PUBMED Abstract]

21. Wong ML, Cooper BA, Paul SM, et al.: Differences in Symptom Clusters Identified Using Ratings of Symptom Occurrence vs. Severity in Lung Cancer Patients Receiving Chemotherapy. J Pain Symptom Manage 54 (2): 194-203, 2017. [PUBMED Abstract]

22. Matzka M, Köck-Hódi S, Jahn P, et al.: Relationship among symptom clusters, quality of life, and treatment-specific optimism in patients with cancer. Support Care Cancer 26 (8): 2685-2693, 2018. [PUBMED Abstract]

23. Papachristou N, Barnaghi P, Cooper B, et al.: Network Analysis of the Multidimensional Symptom Experience of Oncology. Sci Rep 9 (1): 2258, 2019. [PUBMED Abstract]

24. Harris CS, Kober KM, Conley YP, et al.: Symptom clusters in patients receiving chemotherapy: A systematic review. BMJ Support Palliat Care 12 (1): 10-21, 2022. [PUBMED Abstract]

25. Langegård U, Johansson B, Bjork-Eriksson T, et al.: Symptom Clusters in Patients With Brain Tumors Undergoing Proton Beam Therapy. Oncol Nurs Forum 46 (3): 349-363, 2019. [PUBMED Abstract]

26. Chow S, Wan BA, Pidduck W, et al.: Symptom clusters in patients with breast cancer receiving radiation therapy. Eur J Oncol Nurs 42: 14-20, 2019. [PUBMED Abstract]

27. Harris C, Kober KM, Paul SM, et al.: Neurotransmitter Gene Polymorphisms Are Associated with Symptom Clusters in Patients Undergoing Radiation Therapy. Semin Oncol Nurs 39 (4): 151461, 2023. [PUBMED Abstract]

28. Poon M, Dennis K, DeAngelis C, et al.: Symptom clusters of gastrointestinal cancer patients undergoing radiotherapy using the Functional Living Index-Emesis (FLIE) quality-of-life tool. Support Care Cancer 23 (9): 2589-98, 2015. [PUBMED Abstract]

29. Rosenthal DI, Mendoza TR, Fuller CD, et al.: Patterns of symptom burden during radiotherapy or concurrent chemoradiotherapy for head and neck cancer: a prospective analysis using the University of Texas MD Anderson Cancer Center Symptom Inventory-Head and Neck Module. Cancer 120 (13): 1975-84, 2014. [PUBMED Abstract]

30. Xiao W, Chan CWH, Fan Y, et al.: Symptom clusters in patients with nasopharyngeal carcinoma during radiotherapy. Eur J Oncol Nurs 28: 7-13, 2017. [PUBMED Abstract]

31. Kim E, Jahan T, Aouizerat BE, et al.: Changes in symptom clusters in patients undergoing radiation therapy. Support Care Cancer 17 (11): 1383-91, 2009. [PUBMED Abstract]

32. Lemanska A, Dearnaley DP, Jena R, et al.: Older Age, Early Symptoms and Physical Function are Associated with the Severity of Late Symptom Clusters for Men Undergoing Radiotherapy for Prostate Cancer. Clin Oncol (R Coll Radiol) 30 (6): 334-345, 2018. [PUBMED Abstract]

33. Ganesh V, Zhang L, Chan S, et al.: An update in symptom clusters using the Edmonton Symptom Assessment System in a palliative radiotherapy clinic. Support Care Cancer 25 (11): 3321-3327, 2017. [PUBMED Abstract]

34. Ganesh V, Zhang L, Wan BA, et al.: Symptom clusters using the EORTC QLQ-C15-PAL in palliative radiotherapy. Ann Palliat Med 7 (2): 192-204, 2018. [PUBMED Abstract]

35. Chiang SH, Ho KY, Wang SY, et al.: Change in symptom clusters in head and neck cancer patients undergoing postoperative radiotherapy: A longitudinal study. Eur J Oncol Nurs 35: 62-66, 2018. [PUBMED Abstract]

36. Miaskowski C, Conley YP, Mastick J, et al.: Cytokine Gene Polymorphisms Associated With Symptom Clusters in Oncology Patients Undergoing Radiation Therapy. J Pain Symptom Manage 54 (3): 305-316.e3, 2017. [PUBMED Abstract]

37. Harris CS, Miaskowski CA, Conley YP, et al.: Gastrointestinal Symptom Cluster is Associated With Epigenetic Regulation of Lymphotoxin Beta in Oncology Patients Receiving Chemotherapy. Biol Res Nurs 25 (1): 51-64, 2023. [PUBMED Abstract]

Management of Cancer-Related Symptom Clusters

General Approach

The first step in the management of symptom clusters involves an open-ended patient interview to identify symptoms that are bothersome. The next step is to identify the most distressing symptoms with significant negative impact on the patient's quality of life and functioning. A subsequent step is to determine the frequency and severity of the predominant symptoms. This step can be accomplished with a guided interview and objective assessment of individual symptoms using validated questionnaires. For more information, see Fatigue, Depression, and Sleep Disorders.

Another important step is to identify the nature of relationships among the symptoms if one exists. This step is critical to minimize unnecessary interventions that might be burdensome for patients already struggling with multiple symptoms and quality-of-life issues. Associations among symptoms in a cluster inform management of symptoms in that cluster. Different ways that symptoms in a cluster may be related to each other include the following:

- **A primary symptom leading to other symptoms in a symptom cluster** (e.g., insomnia leading to fatigue, anxiety, and depression; hot flashes causing insomnia, which in turn causes anxiety and fatigue; pain causing anxiety and insomnia). In such cases, management will involve first treating the primary symptom if it can be identified.
- **Symptoms as side effects of a treatment used to treat another symptom** (e.g., daytime fatigue as a side effect of hypnotics used to treat insomnia or mirtazapine used to treat depression). Management in such cases involves identifying alternative treatments or optimizing the dose of the triggering medication to minimize the symptom/side effect.
- **Common biological mechanisms underlying multiple symptoms** (e.g., an inflammatory mechanism underlying fatigue, depression, sleep disturbances, and cognitive issues). The underlying inflammatory process may be a treatment target. In some cases, management may involve optimizing cancer treatment regimens, considering the risk-benefit ratio.
- **Common psychological and/or social factors underlying symptom clusters** (e.g., fear of cancer recurrence that causes anxiety, insomnia, and depression; financial or relationship issues as a trigger for insomnia, anxiety, and depression). Addressing the underlying psychological or social factor is the primary treatment strategy in these cases.

Most symptom management studies test the efficacy of the proposed intervention on a single target symptom as the primary outcome and evaluate secondary impact on a related set of symptoms or on the overall symptom burden.[1-4] A small number of studies have evaluated the efficacy of certain interventions on multiple co-occurring and related symptoms in symptom clusters. A randomized controlled trial using cognitive behavioral therapy for insomnia (CBT-I) showed improvement in insomnia, fatigue, and depression symptoms.[1,3,4] Another controlled trial found improvement in cognitive, gastrointestinal, and other symptom clusters using a multimodal intervention that included structured exercise, relaxation, and psychosocial support.[5]

An important part of managing symptom clusters involves patient and caregiver education. Such education includes discussion of co-occurrence of symptoms and their potential relationship to each other. Patients can also be informed that an intervention targeting a specific symptom may impact other, related symptoms. Another step involves identification of symptoms that may be due to side effects of another intervention. In these cases, addressing the target symptom involves optimizing the triggering intervention or, if possible, considering alternative treatments.

If an intervention is deemed necessary, pharmacological and nonpharmacological strategies can be considered (see Table 1).

Table 1. Effective Treatment Modalities Across Multiple Cancer Symptoms and Symptom Clusters

Symptoms	SNRIs[a]	Exercise	E+R	Mindfulness		CBT	
				MBSR-BC	MBAT	CBT	CBT-I
Depression	X	X		X			X
Pain						X	
CIPN	X						
Lymphedema		X					
Anxiety		X		X			
Hot flashes	X						
Fatigue		X		X		X	X

ACT = acceptance and commitment therapy; CBT = cognitive behavioral therapy; CBT-I = cognitive behaviora
= chemotherapy-induced peripheral neuropathy; E+R = exercise + relaxation; MBAT = mindfulness-based art t
mindfulness-based stress reduction–British Columbia; MCP = meaning-centered psychotherapy; PAP = psilocy
psychotherapy; SNRIs = serotonin-norepinephrine reuptake inhibitors.

[a]Duloxetine and venlafaxine.

Symptoms	SNRIs[a]	Exercise	E+R	Mindfulness		CBT	
				MBSR-BC	MBAT	CBT	CBT-I
Low energy				X			
Sleep disturbance		X				X	X
Bowel dysfunction						X	
Perceived stress/distress				X	X	X	
Physical function	X			X			
Psychological/psychosocial function				X		X	
Spiritual well-being							
Quality of life	X				X	X	
Symptom Clusters							
Gastrointestinal (nausea, vomiting, stomach pain, loss of appetite, diarrhea)			X				
Cognitive (decreased concentration, memory problems, fatigue)			X				

ACT = acceptance and commitment therapy; CBT = cognitive behavioral therapy; CBT-I = cognitive behaviora
= chemotherapy-induced peripheral neuropathy; E+R = exercise + relaxation; MBAT = mindfulness-based art t
mindfulness-based stress reduction–British Columbia; MCP = meaning-centered psychotherapy; PAP = psilocy
psychotherapy; SNRIs = serotonin-norepinephrine reuptake inhibitors.

[a]Duloxetine and venlafaxine.

Symptoms	SNRIs[a]	Exercise	E+R	Mindfulness		CBT	
				MBSR-BC	MBAT	CBT	CBT-I
Functional (muscle aches, joint aches)			X				
Mucositis; mouth pain (throat pain, difficulty swallowing)			X				

ACT = acceptance and commitment therapy; CBT = cognitive behavioral therapy; CBT-I = cognitive behaviora
= chemotherapy-induced peripheral neuropathy; E+R = exercise + relaxation; MBAT = mindfulness-based art t
mindfulness-based stress reduction–British Columbia; MCP = meaning-centered psychotherapy; PAP = psiloc
psychotherapy; SNRIs = serotonin-norepinephrine reuptake inhibitors.

[a]Duloxetine and venlafaxine.

For more information about individual side effects, see the summaries on Supportive and Palliative Care.

Pharmacological Interventions

Studies investigating pharmacological interventions generally target a primary symptom while evaluating the impact on other, related symptoms as secondary outcomes. Certain pharmacological interventions have shown efficacy in the management of multiple symptoms in separate studies. For example, serotonin-norepinephrine reuptake inhibitors (SNRIs), specifically duloxetine and venlafaxine, are known to treat depression (see the Pharmacological Intervention section in Depression). SNRIs are also effective in the management of chemotherapy-induced peripheral neuropathy (CIPN) (see the Venlafaxine and duloxetine section in Cancer Pain).[6] Venlafaxine has also shown efficacy in the management of hot flashes in breast cancer survivors.[7] These efficacy studies targeting specific symptoms have been conducted in patients with cancer and in general patient populations.

When pharmacological interventions are considered, the primary goal is to optimize the medication regimen and minimize side-effect burden. Steps to achieve this goal include the following:

- Review the patient's medication list to check if the target symptom is due to side effects of a medication. If a triggering medication is identified, assess if the medication is necessary. If the medication is deemed necessary, examine the dose and/or frequency. Alternatively, consider a substitute medication with similar efficacy but fewer possible side effects causing the target symptom.

- If a pharmacological intervention is considered, it is important to select a medication that can target multiple symptoms if possible. This strategy can help minimize the side-effect burden.

Nonpharmacological Interventions

Nonpharmacological interventions for the management of cancer-related symptom clusters may include exercise, mindfulness, and psychological therapies.

Exercise

Exercise has been defined as physical activity that is planned or structured and involves repetitive bodily movements to improve or maintain cardiorespiratory endurance, muscular strength, muscular endurance, flexibility, and body composition. **Physical activity** has been defined as bodily movements produced by skeletal muscles that result in energy expenditure.[8] Physical activity can include all levels of intensity, including light, moderate, moderate-to-vigorous, and vigorous.[9]

The American College of Sports Medicine's published recommendations for exercise in cancer survivors suggest starting slowly and progressing to 150 minutes of moderate-intensity or 75 minutes of vigorous-intensity aerobic exercise per week.[10] This aerobic exercise is accompanied by 20 to 30 minutes of strength training across all major muscle groups 2 to 3 times per week and regular stretching daily. Investigators have examined multiple forms of physical activity and their impact on cancer-related symptoms. This activity includes aerobic, anaerobic, muscle-strengthening, bone-strengthening, balance, and flexibility exercises, as well as yoga, tai chi, and qigong. However, most of this research has focused on a primary symptom versus a symptom cluster. A recent consensus statement regarding specific exercise guidelines for cancer survivors identified strong evidence for individual symptom management with exercise for anxiety, depression, and fatigue.[11] (Strong evidence was defined as having five or more randomized controlled trials with samples sizes of more than150 participants.) In addition, strong evidence was identified for better management of lymphedema, physical function, and health-related quality of life with exercise.[11] Of note, moderate evidence for bone health and sleep was identified, while insufficient evidence was found for cardiotoxicity, CIPN, pain, cognitive function, falls, nausea, sexual function, and treatment tolerance.[11]

The role of exercise in managing symptom clusters is more limited.[12-15] One study used a multimodal intervention of structured exercise, relaxation, and psychosocial support to address symptom clusters.[5] It noted improvement in gastrointestinal (nausea, vomiting, stomach pain, loss of appetite, diarrhea), cognitive (diminished concentration, memory problems, fatigue), functional (muscle and joint aches), and mucositis (mouth pain, throat pain, and difficulty swallowing) symptom clusters in the treatment group, but no improvement in an affective (nervousness, anxiety, and stress) symptom cluster, compared with usual care.

Mindfulness-based stress reduction (MBSR)

Studies investigating mindfulness commonly involve evaluation of its impact on a primary target symptom and on other, related symptoms as secondary outcomes. Studies that examined different

approaches to MBSR–British Columbia (MBSR-BC) have found predominantly strong positive impacts on symptoms clusters, including combinations of the following:[16-26]

- Perceived stress.
- Anxiety.
- Low energy and physical functioning.
- Psychosocial health.
- Fatigue.
- Depression.
- Feelings of helplessness.
- Hostility/anger.
- Fear of cancer recurrence.
- Coping abilities.
- Catastrophizing.

Many of the MBSR interventions involved multiweek programs in which participants were assessed for outcomes at 3-, 6- and 12-month intervals. During this time, survivors participated in mindfulness training and various exercises and were encouraged to continue using these skills on an ongoing basis. Improvement was measured using the following tools and scales:

- Profile of Mood States.
- Functional Assessment of Cancer Therapy–Endocrine Symptoms.
- World Health Organization-5 Well-being Index.
- Pain Intensity Scale.
- Global Severity Index, SCL-90-R.

MBSR-BC consists of three components:

- Educational material related to relaxation, meditation, the mind-body connection, and a healthy lifestyle for survivors.
- Practice of meditation in group meetings and homework assignments.
- Group processes related to barriers to the practice of meditation and supportive group interaction.

Many studies have evaluated mindfulness for improvement in anxiety and stress symptoms during cancer treatment. Patients who received MBSR while undergoing chemotherapy were shown to have reduced scores on an “overall suffering score.”[16-18,21-25] In two of the studies, technology was used to administer the intervention, one online [25] and another via smartphone.[22] These delivery methods provided more opportunities to incorporate mindfulness practices into the treatment of cancer survivors.

In addition, some studies showed that the intervention groups had a lower symptom burden, compared with control groups, including more significant remission of both anxiety and depressive symptoms.[20] Measures of health-related quality of life (as demonstrated by the Functional Assessment of Cancer Therapy–Breast) were also significantly higher in the intervention group than in the control group.

Fatigue is a common complaint of cancer survivors at all stages, and it contributes to deterioration in many, if not all, psychosocial complaints. Integrating MBSR into cancer care can have many positive outcomes, including reduced fatigue.[17,18,22,26,27] Study participants experienced improvements in other psychosocial areas as well. Compared with control-group participants, those receiving MBSR had a statistically significant reduction in the overall suffering score, improvement in the total Hospital Anxiety and Depression Scale score, and improvement in the total Functional Assessment of Chronic Illness Therapy-Spiritual Well-Being score.[28] In a randomized study of 92 patients, the intervention group experienced positive effects of concurrent MBSR on fatigue and coping while starting adjuvant paclitaxel.[26] While the interventions lasted 1 to 2 months in most cases, results at 6 to 12 months of follow-up showed maintained improvement.[17,18,22,26,27]

Another study, of 191 women with documented stress, showed an overall improvement in psychosocial stress and quality-of-life measures after receiving mindfulness-based art therapy (MBAT) versus standard education support groups. Significant improvements were observed immediately after intervention and 6 months later.[29] Patients with self-reported high levels of stress also experienced stress reduction with MBAT, compared with the standard education support groups.

Psychological therapies

Difficulty with sleep initiation and maintenance often co-occur with anxiety, depression, fatigue, and chronic pain.[30] There are a number of complex interactions with psychological disturbances mediating the relationship between pain and fatigue.[31] With these common symptoms, psychological therapies that target, or are known to create downstream effects on, multiple morbidities help patients address these issues through a single psychotherapeutic approach.

Cognitive behavioral therapy (CBT)

CBT specifically targets the patient's maladaptive thoughts and behaviors that interfere with quality of life. CBT is a problem-focused form of therapy that encourages patients to take an active role in changing their thoughts and behaviors. This form of therapy can address several different cognitive, behavioral, and health targets. It can be provided in individual, group, or telehealth formats. Mobile apps based on cognitive behavioral principles have been successful in treating common psychological symptoms and psychological symptom clusters.[32]

In a multisite randomized controlled trial, 131 patients with cancer were randomly assigned to five sessions of standard CBT, five sessions of profile-tailored CBT, or usual care.[33] Assessments occurred at baseline, 1-month follow-up, and 6-month follow-up. Short- and long-term follow-ups suggested that the patients who received the profile-tailored CBT approach showed significant reductions in levels of pain, pain-associated sleep interference, confusion, bowel patterns, distress, functional status, and

mental quality of life, compared with patients in the other two groups. Patients in the standard CBT group showed improvements in the common symptom clusters, compared with patients in the usual care group.

In another study, a cohort of 86 patients with mixed cancer diagnoses were randomly assigned to a 2-week, patient-directed intervention using cognitive behavioral principles or usual care.[34] The cognitive-behavioral intervention showed efficacy in treating a pain, fatigue, and sleep-disturbance symptom cluster. For patients experiencing this common symptom cluster acutely or mildly, a self-directed approach may be useful.

Acceptance and commitment therapy (ACT)

ACT is called a "Third Wave" CBT approach in that there are similarities to CBT, but the goal is to accept emotions and to understand situations that cannot be changed. This approach is considered an emotion-focused form of therapy. It comprises several techniques, including mindfulness and decreasing judgment about one's thoughts and behaviors in any given situation.

Studies have shown strong empirical support for ACT's positive benefits for the psychological symptom cluster (depression, anxiety, distress, quality of life) for patients at all stages of cancer. However, few studies have examined its effect on physical and health outcomes.[35] Those studies have shown no significant change in physical markers (pain, fatigue, sleep) after ACT therapy in patients with cancer. While there have been empirical studies of ACT in a number of cultures, countries, and ethnicities, most of the research has focused on female-specific cancers such as breast and ovarian.[36-39] There is limited empirical research on the applicability of ACT for males with cancer or patients experiencing physical symptoms.[38]

Meaning-centered psychotherapy

This form of psychotherapy, also called existential psychotherapy or meaning-making psychotherapy, is theoretically rooted in Victor Frankl's book, Man's Search for Meaning.[40] Empirical studies have shown that this approach may be useful for patients experiencing both psychological and common symptom clusters, especially in relation to end-of-life and survivorship concerns.[41]

A randomized controlled trial of 253 patients with stage III or IV solid-tumor cancer compared meaning-centered therapy with the active placebo of supportive group therapy. After 8 weekly group sessions, the group receiving meaning-centered therapy had stronger reductions in the psychological symptom clusters, as well as greater increases in spiritual well-being, than the placebo group.[42,43]

Meaning-centered psychotherapy has also been tested using an individual therapy approach. A group of 120 patients were randomly assigned to 7 weeks of individual manualized psychotherapy or therapeutic massage. At the end of treatment, patients who received meaning-centered psychotherapy showed improvement in both psychological-cluster symptoms (quality of life, spiritual well-being), as well as physical-cluster symptoms (physical symptom distress, number of physical symptoms).[44] A follow-up study randomly assigned 321 patients with advanced cancer to one of three arms: individual meaning-making psychotherapy, individual supportive psychotherapy, or enhanced usual care. After 8 weeks of

treatment, patients receiving meaning-making psychotherapy showed significant improvements in the psychological-symptoms cluster, compared with those receiving enhanced usual care. They also had smaller but significant improvements in the physical-symptoms cluster, compared with those receiving supportive psychotherapy.[45]

Specific Evidence-Based Interventions

Managing Cancer and Living Meaningfully (CALM)

CALM is a brief (2–6 sessions), individual, manualized psychotherapy intervention that is designed to address the psychological symptom cluster in patients with advanced cancer. A randomized controlled trial of 305 patients were divided into CALM (n = 151) or usual care (n = 154) groups. At both the immediate and 6-month follow-up points, patients in the CALM group had fewer depressive symptoms, less end-of-life distress, and greater preparation for the end of life, compared with patients in the usual care group. When CALM outcomes were analyzed among patients with moderate anxiety about death, patients in the CALM group showed significantly greater spiritual well-being and lower levels of anxiety and demoralization at 6 months compared with patients in the usual care group. The brief, individual CALM intervention may be most useful for patients experiencing moderate anxiety about death and the psychological symptom cluster than for those with high or low anxiety about death.[46,47]

Cognitive behavioral therapy for insomnia (CBT-I)

CBT-I includes techniques such as stimulus control, sleep restriction, and relaxation training to address the cognitive and behavioral components that contribute to disrupted sleep. While CBT-I focuses on sleep, studies have shown that it has other psychological and physical impacts.[48] Studies of CBT-I in patients without cancer have found benefits with pain management, depression, and disability.[49]

A 2 x 2 randomized controlled trial of 67 cancer survivors assessed the effects of armodafinil and CBT-I on sleep quality, fatigue, and depression. Symptoms were assessed at three time points: before intervention, immediately after the 7-week intervention, and at the 3-month follow-up. Patients who received CBT-I had significantly improved sleep, fatigue, and depression levels, in addition to reduced hypnotic medication use, at both the post intervention and 3-month follow-up. Patients who received the pharmacological intervention experienced no changes to any of these variables. Path analysis showed that improvement in depressive symptoms was mediated by insomnia severity, suggesting that treatment of sleep disruptions with CBT-I can have a clinically meaningful effect on depressive symptoms.[3]

Studies of CBT-I in patients without cancer have found additional benefits for pain management, depression, and disability.[49] There is extremely limited data on the effect of CBT-I on pain outcomes in patients with cancer.

When patients with cancer are treated at the acute stage of insomnia, minimal and digital forms of CBT-I have also been found to be useful in randomized controlled trials. Minimal forms include psychoeducation booklets and three phone consultations with a sleep psychologist.[50] Digital forms

include self-directed apps via computer or smartphone.[51] These studies have found an impact of CBT-I on sleep measures, as well as psychological-cluster symptoms (depression, anxiety, distress).

Psilocybin-assisted psychotherapy

A potential treatment to address the psychological and common symptom clusters is psilocybin-assisted psychotherapy. While more research is needed before it becomes an evidence-based practice, promising early research results have been disseminated widely on a number of news and blog sites, increasing patient interest in this approach to therapy.

In a double-blind randomized controlled trial, 29 patients with advanced cancer and related anxiety and depression received 7 weeks of psychotherapy and a single dose of either psilocybin or niacin. Patients who received the psilocybin plus psychotherapy showed significant reductions in the severity of symptoms in the psychological symptom cluster.[52] A follow-up study of patients who were still alive (n = 15) showed that these improvements were maintained for 3.2 to 4.5 years after the original study.[53] This emerging treatment continues to develop with wider groups of cancer patients and long-term assessments of multisymptom clusters.

References

1. Heckler CE, Garland SN, Peoples AR, et al.: Cognitive behavioral therapy for insomnia, but not armodafinil, improves fatigue in cancer survivors with insomnia: a randomized placebo-controlled trial. Support Care Cancer 24 (5): 2059-2066, 2016. [PUBMED Abstract]
2. Moss EL, Simpson JS, Pelletier G, et al.: An open-label study of the effects of bupropion SR on fatigue, depression and quality of life of mixed-site cancer patients and their partners. Psychooncology 15 (3): 259-67, 2006. [PUBMED Abstract]
3. Peoples AR, Garland SN, Pigeon WR, et al.: Cognitive Behavioral Therapy for Insomnia Reduces Depression in Cancer Survivors. J Clin Sleep Med 15 (1): 129-137, 2019. [PUBMED Abstract]
4. Roscoe JA, Garland SN, Heckler CE, et al.: Randomized placebo-controlled trial of cognitive behavioral therapy and armodafinil for insomnia after cancer treatment. J Clin Oncol 33 (2): 165-71, 2015. [PUBMED Abstract]
5. Jarden M, Nelausen K, Hovgaard D, et al.: The effect of a multimodal intervention on treatment-related symptoms in patients undergoing hematopoietic stem cell transplantation: a randomized controlled trial. J Pain Symptom Manage 38 (2): 174-90, 2009. [PUBMED Abstract]
6. Song SY, Ko YB, Kim H, et al.: Effect of serotonin-norepinephrine reuptake inhibitors for patients with chemotherapy-induced painful peripheral neuropathy: A meta-analysis. Medicine (Baltimore) 99 (1): e18653, 2020. [PUBMED Abstract]
7. Ramaswami R, Villarreal MD, Pitta DM, et al.: Venlafaxine in management of hot flashes in women with breast cancer: a systematic review and meta-analysis. Breast Cancer Res Treat 152 (2): 231-7, 2015. [PUBMED Abstract]
8. Caspersen CJ, Powell KE, Christenson GM: Physical activity, exercise, and physical fitness: definitions and distinctions for health-related research. Public Health Rep 100 (2): 126-31, 1985. [PUBMED

Abstract]

9. 2018 Physical Activity Guidelines Scientific Committee: 2018 Physical Activity Guidelines Advisory Committee Scientific Report. Washington, DC: U.S. Department of Health and Human Services, 2018. Available online.
10. Schmitz KH, Courneya KS, Matthews C, et al.: American College of Sports Medicine roundtable on exercise guidelines for cancer survivors. Med Sci Sports Exerc 42 (7): 1409-26, 2010. [PUBMED Abstract]
11. Campbell KL, Winters-Stone KM, Wiskemann J, et al.: Exercise Guidelines for Cancer Survivors: Consensus Statement from International Multidisciplinary Roundtable. Med Sci Sports Exerc 51 (11): 2375-2390, 2019. [PUBMED Abstract]
12. Zhi WI, Baser RE, Zhi LM, et al.: Yoga for cancer survivors with chemotherapy-induced peripheral neuropathy: Health-related quality of life outcomes. Cancer Med 10 (16): 5456-5465, 2021. [PUBMED Abstract]
13. Carson JW, Carson KM, Porter LS, et al.: Yoga of Awareness program for menopausal symptoms in breast cancer survivors: results from a randomized trial. Support Care Cancer 17 (10): 1301-9, 2009. [PUBMED Abstract]
14. Cho MH, Dodd MJ, Cooper BA, et al.: Comparisons of exercise dose and symptom severity between exercisers and nonexercisers in women during and after cancer treatment. J Pain Symptom Manage 43 (5): 842-54, 2012. [PUBMED Abstract]
15. Dodd MJ, Cho MH, Miaskowski C, et al.: A randomized controlled trial of home-based exercise for cancer-related fatigue in women during and after chemotherapy with or without radiation therapy. Cancer Nurs 33 (4): 245-57, 2010. [PUBMED Abstract]
16. Henderson VP, Massion AO, Clemow L, et al.: A randomized controlled trial of mindfulness-based stress reduction for women with early-stage breast cancer receiving radiotherapy. Integr Cancer Ther 12 (5): 404-13, 2013. [PUBMED Abstract]
17. Hoffman CJ, Ersser SJ, Hopkinson JB, et al.: Effectiveness of mindfulness-based stress reduction in mood, breast- and endocrine-related quality of life, and well-being in stage 0 to III breast cancer: a randomized, controlled trial. J Clin Oncol 30 (12): 1335-42, 2012. [PUBMED Abstract]
18. Johns SA, Brown LF, Beck-Coon K, et al.: Randomized controlled pilot study of mindfulness-based stress reduction for persistently fatigued cancer survivors. Psychooncology 24 (8): 885-93, 2015. [PUBMED Abstract]
19. Lengacher CA, Johnson-Mallard V, Post-White J, et al.: Randomized controlled trial of mindfulness-based stress reduction (MBSR) for survivors of breast cancer. Psychooncology 18 (12): 1261-72, 2009. [PUBMED Abstract]
20. Liu W, Liu J, Ma L, et al.: Effect of mindfulness yoga on anxiety and depression in early breast cancer patients received adjuvant chemotherapy: a randomized clinical trial. J Cancer Res Clin Oncol 148 (9): 2549-2560, 2022. [PUBMED Abstract]

21. Park S, Sato Y, Takita Y, et al.: Mindfulness-Based Cognitive Therapy for Psychological Distress, Fear of Cancer Recurrence, Fatigue, Spiritual Well-Being, and Quality of Life in Patients With Breast Cancer-A Randomized Controlled Trial. J Pain Symptom Manage 60 (2): 381-389, 2020. [PUBMED Abstract]

22. Shao D, Zhang H, Cui N, et al.: The efficacy and mechanisms of a guided self-help intervention based on mindfulness in patients with breast cancer: A randomized controlled trial. Cancer 127 (9): 1377-1386, 2021. [PUBMED Abstract]

23. Sheikhzadeh M, Zanjani Z, Baari A: Efficacy of Mindfulness-Based Cognitive Therapy and Cognitive Behavioral Therapy for Anxiety, Depression, and Fatigue in Cancer Patients: A Randomized Clinical Trial. Iran J Psychiatry 16 (3): 271-280, 2021. [PUBMED Abstract]

24. Würtzen H, Dalton SO, Elsass P, et al.: Mindfulness significantly reduces self-reported levels of anxiety and depression: results of a randomised controlled trial among 336 Danish women treated for stage I-III breast cancer. Eur J Cancer 49 (6): 1365-73, 2013. [PUBMED Abstract]

25. Yıldırım D, Çiriş Yıldız C: The Effect of Mindfulness-Based Breathing and Music Therapy Practice on Nurses' Stress, Work-Related Strain, and Psychological Well-being During the COVID-19 Pandemic: A Randomized Controlled Trial. Holist Nurs Pract 36 (3): 156-165, 2022 May-Jun 01. [PUBMED Abstract]

26. Gok Metin Z, Karadas C, Izgu N, et al.: Effects of progressive muscle relaxation and mindfulness meditation on fatigue, coping styles, and quality of life in early breast cancer patients: An assessor blinded, three-arm, randomized controlled trial. Eur J Oncol Nurs 42: 116-125, 2019. [PUBMED Abstract]

27. Salvador C, Mark P, Hoenemeyer T, et al.: Prospective feasibility study of a mindfulness-based program for breast cancer patients in the southeastern US. Complement Ther Clin Pract 49: 101639, 2022. [PUBMED Abstract]

28. Tan SB, Chee CH, Ngai CF, et al.: Mindfulness-based supportive therapy on reducing suffering in patients with advanced cancer: randomised controlled trial. BMJ Support Palliat Care 13 (e3): e1117-e1125, 2024. [PUBMED Abstract]

29. Marzorati C, Riva S, Pravettoni G: Who Is a Cancer Survivor? A Systematic Review of Published Definitions. J Cancer Educ 32 (2): 228-237, 2017. [PUBMED Abstract]

30. Calvo-Schimmel A, Paul SM, Cooper BA, et al.: Oncology Outpatients With Worse Anxiety and Sleep Disturbance Profiles Are at Increased Risk for a Higher Symptom Burden and Poorer Quality of Life. Cancer Nurs 46 (6): 417-431, 2023 Nov-Dec 01. [PUBMED Abstract]

31. Bonhof CS, van de Poll-Franse LV, Vissers PAJ, et al.: Anxiety and depression mediate the association between chemotherapy-induced peripheral neuropathy and fatigue: Results from the population-based PROFILES registry. Psychooncology 28 (9): 1926-1933, 2019. [PUBMED Abstract]

32. Kwekkeboom KL, Abbott-Anderson K, Wanta B: Feasibility of a patient-controlled cognitive-behavioral intervention for pain, fatigue, and sleep disturbance in cancer. Oncol Nurs Forum 37 (3): E151-9, 2010. [PUBMED Abstract]

33. Dalton JA, Keefe FJ, Carlson J, et al.: Tailoring cognitive-behavioral treatment for cancer pain. Pain Manag Nurs 5 (1): 3-18, 2004. [PUBMED Abstract]

34. Kwekkeboom KL, Abbott-Anderson K, Cherwin C, et al.: Pilot randomized controlled trial of a patient-controlled cognitive-behavioral intervention for the pain, fatigue, and sleep disturbance symptom cluster in cancer. J Pain Symptom Manage 44 (6): 810-22, 2012. [PUBMED Abstract]

35. Zhao C, Lai L, Zhang L, et al.: The effects of acceptance and commitment therapy on the psychological and physical outcomes among cancer patients: A meta-analysis with trial sequential analysis. J Psychosom Res 140: 110304, 2021. [PUBMED Abstract]

36. Najvani BD, Neshatdoost HT, Abedi MR: The effect of acceptance and commitment therapy on depression and psychological flexibility in women with breast cancer. Zahedan J Res Med Sci. 17 (4): e965, 2015. Available online.

37. Montazer A, Nemati F, Dehghani F: Efficacy of acceptance and commitment therapy on breast cancer female patients' hope. Iranian Journal of Cancer Prevention. 10 (2): e5526, 2017. Available online.

38. Mosher CE, Secinti E, Li R, et al.: Acceptance and commitment therapy for symptom interference in metastatic breast cancer patients: a pilot randomized trial. Support Care Cancer 26 (6): 1993-2004, 2018. [PUBMED Abstract]

39. Rost AD, Wilson K, Buchanan E: Improving psychological adjustment among late-stage ovarian cancer patients: examining the role of avoidance in treatment. Cognitive and Behavioral Practice. 19 (4): 508-17, 2012. Available online.

40. Frankl VE: Man's Search for Meaning. Beacon Press, 2006. Also available online.

41. van der Spek N, Vos J, van Uden-Kraan CF, et al.: Meaning making in cancer survivors: a focus group study. PLoS One 8 (9): e76089, 2013. [PUBMED Abstract]

42. Breitbart W, Rosenfeld B, Gibson C, et al.: Meaning-centered group psychotherapy for patients with advanced cancer: a pilot randomized controlled trial. Psychooncology 19 (1): 21-8, 2010. [PUBMED Abstract]

43. Breitbart W, Rosenfeld B, Pessin H, et al.: Meaning-centered group psychotherapy: an effective intervention for improving psychological well-being in patients with advanced cancer. J Clin Oncol 33 (7): 749-54, 2015. [PUBMED Abstract]

44. Breitbart W, Poppito S, Rosenfeld B, et al.: Pilot randomized controlled trial of individual meaning-centered psychotherapy for patients with advanced cancer. J Clin Oncol 30 (12): 1304-9, 2012. [PUBMED Abstract]

45. Breitbart W, Pessin H, Rosenfeld B, et al.: Individual meaning-centered psychotherapy for the treatment of psychological and existential distress: A randomized controlled trial in patients with advanced cancer. Cancer 124 (15): 3231-3239, 2018. [PUBMED Abstract]

46. Lo C, Hales S, Jung J, et al.: Managing Cancer And Living Meaningfully (CALM): phase 2 trial of a brief individual psychotherapy for patients with advanced cancer. Palliat Med 28 (3): 234-42, 2014. [PUBMED Abstract]

47. Rodin G, Lo C, Rydall A, et al.: Managing Cancer and Living Meaningfully (CALM): A Randomized Controlled Trial of a Psychological Intervention for Patients With Advanced Cancer. J Clin Oncol 36 (23): 2422-2432, 2018. [PUBMED Abstract]
48. Matthews EE, Berger AM, Schmiege SJ, et al.: Cognitive behavioral therapy for insomnia outcomes in women after primary breast cancer treatment: a randomized, controlled trial. Oncol Nurs Forum 41 (3): 241-53, 2014. [PUBMED Abstract]
49. Enomoto K, Adachi T, Fujino H, et al.: Comparison of the effectiveness of cognitive behavioral therapy for insomnia, cognitive behavioral therapy for pain, and hybrid cognitive behavioral therapy for insomnia and pain in individuals with comorbid insomnia and chronic pain: A systematic review and network meta-analysis. Sleep Med Rev 66: 101693, 2022. [PUBMED Abstract]
50. Casault L, Savard J, Ivers H, et al.: A randomized-controlled trial of an early minimal cognitive-behavioural therapy for insomnia comorbid with cancer. Behav Res Ther 67: 45-54, 2015. [PUBMED Abstract]
51. Ritterband LM, Bailey ET, Thorndike FP, et al.: Initial evaluation of an Internet intervention to improve the sleep of cancer survivors with insomnia. Psychooncology 21 (7): 695-705, 2012. [PUBMED Abstract]
52. Ross S, Bossis A, Guss J, et al.: Rapid and sustained symptom reduction following psilocybin treatment for anxiety and depression in patients with life-threatening cancer: a randomized controlled trial. J Psychopharmacol 30 (12): 1165-1180, 2016. [PUBMED Abstract]
53. Agin-Liebes GI, Malone T, Yalch MM, et al.: Long-term follow-up of psilocybin-assisted psychotherapy for psychiatric and existential distress in patients with life-threatening cancer. J Psychopharmacol 34 (2): 155-166, 2020. [PUBMED Abstract]

Latest Updates to This Summary (07/11/2024)

The PDQ cancer information summaries are reviewed regularly and updated as new information becomes available. This section describes the latest changes made to this summary as of the date above.

Editorial changes were made to this summary.

This is a new summary.

This summary is written and maintained by the PDQ Supportive and Palliative Care Editorial Board, which is editorially independent of NCI. The summary reflects an independent review of the literature and does not represent a policy statement of NCI or NIH. More information about summary policies and the role of the PDQ Editorial Boards in maintaining the PDQ summaries can be found on the About This PDQ Summary and PDQ® Cancer Information for Health Professionals pages.

About This PDQ Summary

Purpose of This Summary

This PDQ cancer information summary for health professionals provides comprehensive, peer-reviewed, evidence-based information about the identification and management of symptom clusters. It is intended as a resource to inform and assist clinicians in the care of their patients. It does not provide formal guidelines or recommendations for making health care decisions.

Reviewers and Updates

This summary is reviewed regularly and updated as necessary by the PDQ Supportive and Palliative Care Editorial Board, which is editorially independent of the National Cancer Institute (NCI). The summary reflects an independent review of the literature and does not represent a policy statement of NCI or the National Institutes of Health (NIH).

Board members review recently published articles each month to determine whether an article should:

- be discussed at a meeting,
- be cited with text, or
- replace or update an existing article that is already cited.

Changes to the summaries are made through a consensus process in which Board members evaluate the strength of the evidence in the published articles and determine how the article should be included in the summary.

The lead reviewers for Symptom Clusters in Cancer are:

- Marilyn J. Hammer, PhD, DC, RN, FAAN (Dana-Farber Cancer Institute)
- Jayesh Kamath, MD, PhD (University of Connecticut Health Center)
- Edward B. Perry, MD (VA Connecticut Healthcare System)
- Rachel A. Pozzar, PhD, RN (Dana-Farber Cancer Institute)
- Diane Von Ah, PhD, RN, FAAN (The Ohio State University)
- Amy Wachholtz, PhD, MDiv, MS, ABPP (University of Colorado)

Any comments or questions about the summary content should be submitted to Cancer.gov through the NCI website's Email Us. Do not contact the individual Board Members with questions or comments about the summaries. Board members will not respond to individual inquiries.

Levels of Evidence

Some of the reference citations in this summary are accompanied by a level-of-evidence designation. These designations are intended to help readers assess the strength of the evidence supporting the use

of specific interventions or approaches. The PDQ Supportive and Palliative Care Editorial Board uses a formal evidence ranking system in developing its level-of-evidence designations.

Permission to Use This Summary

PDQ is a registered trademark. Although the content of PDQ documents can be used freely as text, it cannot be identified as an NCI PDQ cancer information summary unless it is presented in its entirety and is regularly updated. However, an author would be permitted to write a sentence such as "NCI's PDQ cancer information summary about breast cancer prevention states the risks succinctly: [include excerpt from the summary]."

The preferred citation for this PDQ summary is:

PDQ® Supportive and Palliative Care Editorial Board. PDQ Symptom Clusters in Cancer. Bethesda, MD: National Cancer Institute. Updated <MM/DD/YYYY>. Available at: https://www.cancer.gov/about-cancer/treatment/side-effects/cancer-symptom-clusters-hp-pdq. Accessed <MM/DD/YYYY>.

Images in this summary are used with permission of the author(s), artist, and/or publisher for use within the PDQ summaries only. Permission to use images outside the context of PDQ information must be obtained from the owner(s) and cannot be granted by the National Cancer Institute. Information about using the illustrations in this summary, along with many other cancer-related images, is available in Visuals Online, a collection of over 2,000 scientific images.

Disclaimer

The information in these summaries should not be used as a basis for insurance reimbursement determinations. More information on insurance coverage is available on Cancer.gov on the Managing Cancer Care page.

Contact Us

More information about contacting us or receiving help with the Cancer.gov website can be found on our Contact Us for Help page. Questions can also be submitted to Cancer.gov through the website's Email Us.

Updated: July 11, 2024

If you would like to reproduce some or all of this content, see Reuse of NCI Information for guidance about copyright and permissions. In the case of permitted digital reproduction, please credit the National Cancer Institute as the source and link to the original NCI product using the original product's title; e.g., "Symptom Clusters in Cancer (PDQ®)–Health Professional Version was originally published by the National Cancer Institute."